"[d]igital innovations and efficiencies since 2005 have undoubtedly reduced the costs of collecting, storing, processing, and transmitting information" and that "despite reduced costs, increased efficiency, and all the new data and computing power available, the access fee cap has remained fixed at an inflated level that reflects the technology capabilities of 2005."[365]

How to Improve Market Structure, Curt Bradbury, Chief Operating Officer, Stephen's Inc. and Kenneth E. Bentsen, Jr., President and CEO, SIFMA (SIFMA Market Structure Task Force Recommendations), dated July 14, 2014 (stating "Access fees charged by exchanges and other venues should be dramatically reduced, if not eliminated. While brokers are legally required to route their orders to the exchange that is quoting the best price – so called 'protected quotes' – the exchanges are permitted to charge relatively high fees for accessing these quotes: currently 30 cents for every 100 shares. These fees have distorted market pricing as they are a significant percentage of overall trading costs and are several times higher than the fees charged by off-exchange venues. As a result, brokers often avoid routing their orders to exchanges. Exchanges also rebate most of their access fee revenue through price structures such as 'maker/taker.' These developments have led to a proliferation of order types designed to avoid access fees and capture rebates, and that proliferation, in turn, adds complexity to the system, requires continuing technology changes and creates potential for market instability. The Securities and Exchange Commission should reduce the current cap on access fees to no more than 5 cents per 100 shares, and indeed should consider eliminating access fees altogether."); Bradley Hope & Scott Patterson, "NYSE Plan Would Revamp Trading," WALL ST. J. (Dec. 17, 2014), available at http://www.wsj.com/articles/intercontinental-exchangeproposing-major-stock-market-overhaul-1418844900 (stating that since 2005, when the fee cap of $0.003 per share was chosen, competitive and technological advancements have led to decreased costs (spreads and commissions), and as a result, access fees have become a larger portion of overall transaction costs); ICE's Six Recommendations for Reforming Markets," WALL ST. J. (Dec. 18, 2014), available at http://blogs.wsj.com/moneybeat/2014/12/18/ices-six-recommendations-for-reformingmarkets/ (recommending reduction in the access fee cap to $0.0005 in conjunction with adoption of a "trade at" rule, "eliminating maker-taker pricing" and stating "[w]ith myriad different make-take and take-make pricing models in existence today, we believe the potential conflicts and complexity that ensue from the maker-taker models outweigh any perceived benefits. We believe there are better options available to incentivize market-makers to maintain two-sided quotes and reduce intraday volatility" including incentive programs that would obligate market makers to provide liquidity); Joe Ratterman, Chief Executive Officer, & Chris Concannon, President, BATS, "Open Letter to U.S. Securities Industry Participants Re: Market Structure Reform Discussion," at 1 (Jan. 6, 2015), available at http://cdn.batstrading.com/resources/newsletters/OpenLetter010615.pdf (stating access fee cap "requires a substantial reduction and restructuring" and further stating the cap "has remained unchanged for far too long and has never been reevaluated for potential market distortions given the substantially altered broker models and reductions in commissions since the implementation of Regulation NMS.").

365 See IEX Letter IV at 8. See also e.g., Letter from Paul M. Russo, Managing Director, Goldman Sachs & Co. LLC, to Brent J. Fields, Secretary, Commission, at 3 (May 24, 2018), available at https://www.sec.gov/comments/s7-05-18/s70518-3711788-162473.pdf ("Goldman 2018 Letter") (commenting on File No. S7-05-18 "Transaction Fee Pilot for NMS Stocks") at 2 (stating in 2018 "In the thirteen years since the Commission adopted the Fee Cap, spreads have considerably narrowed and commission rates have contracted. However, the Fee Cap has remained unadjusted. There is a well-developed, general consensus among market participants that a [30 mil] per share Fee Cap is an outdated benchmark for execution costs in today's trading environment. As a limit, it creates an upper-range that is simply too high and far from representative of true prices in the marketplace.").

1. Amendments to Rule 612

As discussed above, the Commission is adopting a smaller minimum pricing increment of \$0.005 for certain NMS stocks. Because the Commission is adopting the smaller \$0.005 increment, it is also amending the preexisting access fee caps in Rule 610(c) to prevent introducing new pricing distortions in the market.[366] For protected quotations priced \$1.00 or more, the Commission is adopting a 10 mil access fee cap and has decided that this level is appropriate based on several additional considerations, as discussed in the following sections.

2. Exchange Fee Models

The exchange fee structures in the national market system that have developed under Rule 610 are complex and consist of fees charged and rebates paid to market participants. As stated above, exchanges using a "maker-taker" pricing model, pay a rebate to a "maker" or provider of liquidity, which is funded by the fees charged to a "taker" of liquidity.[367] The exchange earns as revenue the difference between the fee paid by the taker and the rebate paid to the provider or maker.[368] For maker-taker exchanges, the amount of the taker fee is limited by

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366 See infra section VII.D.2.a.

367 See SRO fee schedules, which are available on each SRO's website. See also infra section VII.C.2.c, Table 4. This discussion focuses on exchange fees because, currently, exchanges are the only trading centers that display protected quotations. If an ATS or OTC market maker displayed a protected quotation, its fees would be subject to the access fee caps under Rule 610(c).

368 A few exchanges have adopted a "taker-maker" pricing model (also called an inverted model), in which they charge a fee to the provider of liquidity and pay a rebate to the taker of liquidity. See, e.g., Nasdaq BX fee schedule available at http://www.nasdaqtrader.com/trader.aspx?id=bx_pricing (as of Feb. 2024); NYSE National fee schedule available at https://www.nyse.com/publicdocs/nyse/regulation/nyse/NYSE_National_Schedule_of_Fees.pdf (as of Jan. 1, 2024); Cboe BYX fee schedule available at https://www.cboe.com/us/equities/membership/fee_schedule/byx/ (as of Feb. 2024); and Cboe EDGA fee schedule available at https://www.cboe.com/us/equities/membership/fee_schedule/edga/ (as of Feb., 2024). See also infra section VII.C.2.c, Table 4. For taker-maker exchanges, the amount of the maker fee charged to the provider of liquidity is not bounded by the Rule 610(c) access fee cap because such fee is not a charge to access the market's best bid/offer for NMS stocks, but such fees typically are no more than \$0.0030.

the access fee caps imposed by Rule 610(c). The Rule 610(c) access fee caps apply to the fees assessed on an incoming order that executes against a resting protected quote, but do not apply to the rebates. However, the Rule 610(c) access fee caps indirectly limit the average amount of the rebates that an exchange offers to about $0.0030 per share in order to maintain net positive transaction revenues. Thus, an exchange may charge higher access fees to fund higher liquidity rebates.[369] Some exchanges state that rebates are necessary in order for them to attract trading volume.[370]

In recent years, a variety of concerns have been stated about the prevailing maker-taker fee model, and particularly the rebates paid by the exchanges. Those include that the fee/rebate models: (1) undermine market transparency since displayed prices do not account for exchange transaction fees or rebates and therefore do not reflect the net economic costs of a trade;[371] (2) serve as a way to effectively quote in sub-penny increments on a net basis when the effect of a maker-taker exchange's sub-penny rebate is taken into account even though the minimum quoting increment is expressed in full pennies;[372] (3) introduce unnecessary market complexity through the proliferation of new exchange order types (and new exchanges) designed solely to

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369 This was one of the concerns the Commission identified when it approved the access fee caps. See Regulation NMS Adopting Release, supra note 4, at 37545. ("[T]he fee limitation is necessary to achieve the purposes of the Exchange Act. Access fees tend to be highest when markets use them to fund substantial rebates to liquidity providers, rather than merely to compensate for agency services.").

370 See, e.g., Letter from Patrick Sexton, EVP, General Counsel & Corporate Secretary, Cboe Global Markets, Inc., dated Apr. 5, 2024 ("Cboe Letter IV") at 2-5; Letter from Brett Kitt, Vice President, Deputy General Counsel, Nasdaq, Inc., dated Mar. 25, 2024 ("Nasdaq Letter IV") at 3-5; Nasdaq Letter III at 2-5.

371 See, e.g., Letter from Richard Steiner, Global Equities Liaison to Regulatory & Government Affairs, RBC Capital Markets, to Elizabeth Murphy, Secretary, Commission, at 2-3 (Nov. 22, 2013), available at https://www.sec.gov/comments/s7-02-10/s70210-411.pdf ("RBC Capital Letter") (commenting on potential equity market structure initiatives).

372 See, e.g., Larry Harris, "Maker-Taker Pricing Effects on Market Quotations," at 24-25 (Nov. 14, 2013).

take advantage of pricing models;[373] (4) drive orders to non-exchange trading centers that do not display quotes as market participants seek to avoid the higher fees that exchanges charge to subsidize the rebates they offer to attract liquidity;[374] and (5) benefit sophisticated market participants like market makers and proprietary traders at the expense of other market participants.[375] Further, the prevailing access fee structure creates potential conflicts of interest for broker-dealers, who must provide the best execution to their customers' orders while facing potentially conflicting economic incentives to avoid fees or earn rebates from the trading centers to which they direct those orders for execution.[376]

a. Transparency

The chart below illustrates with a hypothetical example the Commission's concern that exchange fee/rebate models can undermine price transparency. While some investors may invest

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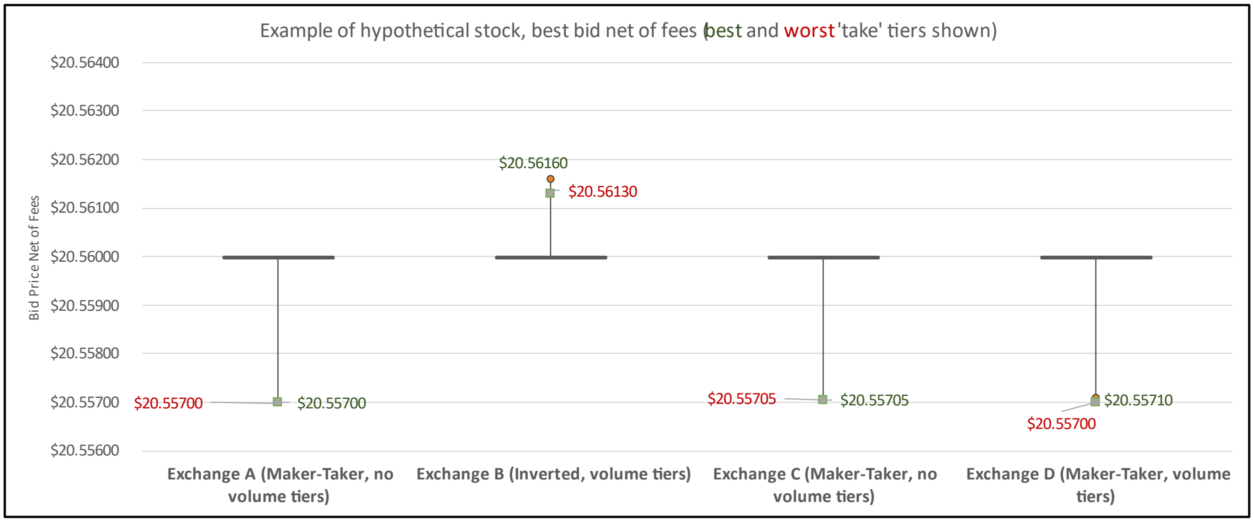
373 See, e.g., Curt Bradbury, Market Structure Task Force Chair, Board of Directors, SIFMA, and Kenneth E. Bentsen Jr., President and Chief Executive Officer, SIFMA, Opinion, "How to Improve Market Structure," N.Y. Times (July 14, 2014), available at http://dealbook.nytimes.com/2014/07/14/how-to-improve-market-structure/?_r=0 (stating that the "proliferation of order types designed to avoid access fees and capture rebates . . . adds complexity to the system, requires continuing technology changes and creates potential for market instability" and recommending access fees charged by exchanges be "dramatically reduced, if not eliminated"); RBC Capital Letter at 2; and Letter from Haim Bodek, Managing Principal and Stanislav Dolgopolov, Regulatory Consultant, Decimus Capital Markets, LLC, dated Apr. 25, 2016, at 3 and 11 ("Decimus 2016 Letter"); Vanguard Letter at 6.

374 See, e.g., Menkveld, Albert J., Bart Zhou Yueshen, and Haoxiang Zhu, "Shades of darkness: A pecking order of trading venues." *Journal of Financial Economics* 124, no. 3 (2017) at 503-534, available at https://www.mit.edu/~zhuh/MenkveldYueshenZhu_2017JFE_dark.pdf; RBC Capital Letter at 2.

375 See, e.g., RBC Capital Letter at 2-4; Letter from Mehmet Kinak, Vice President – Global Head of Systematic Trading & Market Structure, and Jonathan Siegel, Vice President – Senior Legal Counsel (Legislative & Regulatory Affairs), T. Rowe Price, to Brent J. Fields, Secretary, Commission, dated June 12, 2018, at 2, available at https://www.sec.gov/comments/s7-05-18/s70518-3832746-162769.pdf (sec.gov) (commenting on File No. S7-05-18 "Transaction Fee Pilot for NMS Stocks).

376 See, e.g., Stanislav Dolgopolov, "The Maker-Taker Pricing Model and its Impact on the Securities Market Structure: A Can of Worms for Securities Fraud?" 8 VA. L. & BUS. REV. 231, 270 (2014), available at http://papers.ssrn.com/sol3/papers.cfm?abstract_id=2399821 (retrieved from SSRN Elsevier database).

heavily to fully map out the fee schedules, these schedules are complex and thus doing so would be costly, consequently it is likely that not all investors fully map out fee schedules.[377]



The Commission examined data for NMS stocks to demonstrate the price variations that can occur under the exchange fee and rebate schedules. The chart above shows a common scenario for a hypothetical stock, with four exchanges all displaying the same best bid of $20.56. However, due to differing fee schedules, each of the four represents a different net bid to market participants. Furthermore, due to volume-based price fee tiers, some exchanges' net bids differ for different market participants. The fee schedules used to create this example are taken from various exchange fee schedules as of June 2024. Exchange A charges the maximum allowed access fee of 30 mils, and so market participants that trade with the Exchange A bid receive a net price of $20.557 per share ($20.56 minus a $0.003 fee). Exchange B has an inverted "taker-maker" fee schedule and so pays a rebate to participants who remove liquidity, with differing

377 See infra section VII.C.2.c and Table 4 therein for additional analysis and discussion of the complexity of fee and rebate schedules.

rebates based on volume tiers. In this example, a market participant that trades with the bid on Exchange B receives a net price of \$20.5616 if in the best tier (\$20.56 plus a \$0.0016 rebate). Exchange C charges an access fee of 29.5 mils, meaning that a market participant that trades with the Exchange C bid would receive a net price of \$20.55705 per share (\$20.56 minus the \$0.00295 fee). Lastly, Exchange D charges an access fee of 29 mils to those in the best tier, so such a participant trading with Exchange D receives a net price of \$20.5571. Despite all four exchanges showing the same bid of \$20.56, the bids net of fees vary from a low of \$20.557 to a high of \$20.5616, a substantial difference of \$0.0046 per share (nearly half the current minimum pricing increment).

b. Liquidity and the NBBO

The price of liquidity for investors in terms of buying and then later selling a security is the spread between the best bid and the best offer, which is reflected by the NBBO. For those market participants that provide liquidity, such as market makers, this spread similarly represents the market price for providing those liquidity services at any given point in time. More recently, trading center models that pay rebates to liquidity providers (which rebates are funded on a transaction basis by charging an access fee to the taker of liquidity) pay an additional return to the liquidity provider separate from what would be captured though the spread, which may then lead those liquidity providers to lower the spread (that is, the implicit price for their liquidity provision services) more than they would otherwise.[378] These prices are reflected in the NBBO, which is disseminated in the national market system.

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[378] See also infra section VII.B.3.

Others have stated that the maker-taker model has positive effects by enabling exchanges to compete with non-exchange trading centers and by narrowing quoted spreads through subsidizing posted prices,[379] but these potential benefits should be balanced against the market distortions associated with the fee and rebate models mentioned above.[380] Further, rebates paid to liquidity providers under maker-taker fee schedules may narrow displayed spreads in some securities by subsidizing liquidity providers (i.e., by allowing a maker to post a more aggressive price than it may have in absence of a rebate), and these prices may not reflect the underlying economics for the NMS stock.[381] In turn, that displayed liquidity may establish the NBBO,[382] which is often used as the benchmark for marketable order flow, including retail order flow, that is executed off-exchange by either matching or improving upon those distortive prices.[383] Accordingly, rebates may distort quotation prices that are displayed in the national market system.[384]

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379 See, e.g., Letter from John A. Zecca, Executive Vice President, Global Chief Legal, Risk and Regulatory Officer, Nasdaq, Inc., dated Aug. 9, 2023 ("Nasdaq Letter II") at 5-6; Larry Harris, "Maker-Taker Pricing Effects on Market Quotations," at 5 (Nov. 14, 2013), available at https://en-coller.tau.ac.il/sites/nihul_en.tau.ac.il/files/media_server/Recanati/management/seminars/account/Maker.pdf; Letter from Richie Prager, Managing Director, Head of Trading and Liquidity Strategies, BlackRock, Inc., to Mary Jo White, Chair, SEC, at 2 (Sept. 12, 2014), available at https://www.sec.gov/comments/s7-02-10/s70210-419.pdf; Michael Brolley & Katya Malinova, "Informed Trading and Maker-Taker Fees in a Low Latency Limit Order Market," at 2 (Oct. 24, 2013), available at http://papers.ssrn.com/sol3/papers.cfm?abstract_id=2178102.

380 As discussed throughout, these concerns include, for example, undermining price transparency, introducing unnecessary complexity through the proliferation of new exchange order types and order routing strategies, added market fragmentation, creating or exacerbating potential conflicts of interest between brokers and their customers, and in NMS stocks that are tick constrained, assessing a higher cost to liquidity demanders.

381 See also infra sections VII.B.3 and VII.C.2.

382 See also infra sections VII.B.3 and VII.C.2.

383 See, e.g., Concept Release on Equity Market Structure, supra note 59 (evaluating broadly the performance of market structure since Regulation NMS, particularly for long-term investors and for businesses seeking to raise capital, and soliciting comment on whether regulatory initiatives to improve market structure are needed).

384 See infra section VII.A and infra notes 1733 and 1734 and accompanying text.

High rebates also incentivize excessive intermediation,[385] especially in very liquid securities, to the extent rebates induce or exacerbate an oversupply of liquidity at the best bid or offer.[386] As discussed below, some stocks will trade with a quoted spread one tick wide, whether the tick size is $0.01 or $0.005. In those cases, quoted spread is unable to adjust lower due to the tick size, so the rebate paid to liquidity providers, which is funded by an access fee paid by liquidity takers, acts as a wealth transfer from liquidity takers to liquidity providers. This wealth transfer in turn unnecessarily incentivizes the provision of liquidity (i.e., encourages providing liquidity in order to earn the rebate), thereby creating an environment with too much liquidity supplied relative to liquidity demanded and leading to rebates for faster liquidity suppliers and a higher cost to liquidity takers.[387] In other words, excessive quoting in tick-constrained securities to earn rebates undermines price transparency because displayed prices and the associated size at those prices do not reflect the underlying economics of supply and demand, but rather reflect the impact of fees or rebates.[388]

c. Potential Conflicts of Interest

As more fully discussed below and in the Economic Analysis, access fees and rebates create potential broker-dealer conflicts of interest in order routing, particularly to the extent the fees and rebates are not passed through to the customer.[389] For example, this structure can create an incentive for a broker-dealer that fully absorbs transaction costs or rebates potentially to route customer orders to an exchange in order to avoid fees that are paid by the broker-dealer or to

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385 See infra note 1005.

386 See Proposing Release, supra note 11, at 80292. See also infra note 1005 and accompanying text.

387 See infra section VII.D.2.

388 See infra note 1005 and accompanying text.

389 See infra sections IV.D.1.b and VII.D.2.c. See also Proposing Release, supra note 11, at 80330.

receive the highest rebate paid.[390] Lowering the access fees caps will help alleviate potential conflicts of interest.[391]

d. Market Complexity

Exchange fee and rebate models under preexisting Rule 610(c) have also led to the development of a variety of complex order types, including those that allow market participants to avoid paying access fees or to ensure that rebates are collected.[392] Further, exchange fee and rebate models are another way exchanges differentiate themselves from each other and from other trading centers to attract order flow. The variability of fee and rebate models (often within the same exchange group) introduces additional market complexity and fragmentation because it encourages the creation of new exchanges that offer different pricing structures to attract different types of market participants or trading strategies. Moreover, the drive to establish novel and competitive fee schedules results in frequent fee schedule changes (typically on a monthly basis), which adds uncertainty and complexity to the marketplace because market participants must continually update their routing tables to reflect these price changes.[393] A higher cap allows for a wider range of possible access fees (i.e., \$0-\$0.0030) and more variability in exchange fees, which introduces additional complexity to the market.[394]

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390 See, e.g., Vanguard Letter at 6.

391 See infra section VII.D.2.d.

392 Examples of such complex order types include post-only orders or add-liquidity only orders that seek to only provide liquidity to gain a rebate and will not upon entry, execute against a resting order on the other side of the market so as to avoid paying a transaction fee. See, e.g., BZX Rule 11.9(c)(6) (defining a BZX Post Only Order); MEMX Rule 11.16(i)(6) (defining Post Only); Nasdaq Equity, Rule 4702(4)(A)(defining a Post-Only Order); NYSE Rule 7.31(e)(2) (defining an ALO Order); NYSE Arca Rule 7.31(e)(2) (defining an ALO Order). See also Staff Report on Algorithmic Trading in the U.S. Capital Markets (Aug. 5, 2020), available at https://www.sec.gov/files/Algo_Trading_Report_2020.pdf.

393 See infra note 1081 and accompanying text.

394 See supra note 361 and infra note 1764.

C. Proposal to Amend 610(c)

To accommodate the amendments to Rule 612 and to address the issues that have developed with the fee schedules for protected quotes under Rule 610(c), the Commission proposed to amend Rule 610(c) by reducing the level of the access fee caps for all protected quotations in NMS stocks. For protected quotations in all NMS stocks priced $1.00 or more, the proposal introduced two lower access fee caps to accommodate the proposed minimum pricing increments under proposed Rule 612 for quotations priced equal to or greater than $1.00 per share.[395] Specifically, for all protected quotations in NMS stocks priced $1.00 or more per share and assigned a proposed minimum pricing increment greater than $0.001, the Commission proposed a 10 mil access fee cap; and for all protected quotations in NMS stocks priced $1.00 or more per share and assigned the proposed $0.001 minimum pricing increment, the Commission proposed a 5 mil access fee cap. For all protected quotations in NMS stocks priced less than $1.00 per share, the Commission proposed to reduce the access fee cap to 0.05% of the quotation price.[396]

The Commission received many comments on the proposed amendments to Rule 610(c). The comments are discussed more fully below.

D. Final Rule 610(c)

The amended access fee caps reflect several considerations by the Commission as well as input from commenters. For protected quotations priced $1.00 or more, the Commission is adopting a 10 mil access fee cap. This level is appropriate based on several considerations. First, because the Commission is adopting the $0.005 increment for certain NMS stocks under

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395 See Proposed Rule 610(c); Proposing Release, supra note 11, at 80269.

396 See Proposed Rule 610(c); Proposing Release, supra note 11, at 80269.